LOGICHAIN 2026: THE VEILREIGN

Sustainable Supply Chain in the Garment and Textile Industry

ROUND 2A | DATA CASE STUDY

TEAM: SUPERNOVA

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Appendix: the calculation workbook (Tasks 1 to 5, all formulas live) is submitted separately as PHỤ LỤC and is also available online: PHU LUC | Supernova.xlsx

1. INTRODUCTION

TEXIN is a Vietnamese garment manufacturer in Trang Bang that sews and finishes five sportswear lines: running T-shirts, football jerseys, hiking shorts, training jackets and fitness sweatshirts. Its customer **LSC** is an EU buyer that collects the goods ex-works (EXW) at the factory and ships them through Cat Lai to distribution centres in France, Portugal and Romania. In 2026 TEXIN sold **5.02 million pieces** for **USD 19.28 million** (USD 3.84 per piece) at a gross margin of 20.5%.

Dyeing and finishing run on steam from a **coal-fired boiler** that supplies 300,000 GJ of useful heat a year from about 12,000 t of coal and emits **28,380 tCO2e**, the baseline of the case. LSC now asks its suppliers to measure and cut emissions, move to renewable energy and phase out coal. It offers TEXIN potential orders growing 22% to 28% a year if TEXIN commits to a five-year ESG roadmap, while a supplier that stays on coal is expected to lose 10% to 15% of its orders each year. The roadmap is a biomass conversion (70% rice husk pellet and 30% wood pellet by heat) with a total project CAPEX of **USD 940,500**.

The report follows the five case requirements in Sections 3 to 6, after the executive summary in Section 2. All figures come from the case workbook or from the team workbook submitted as the appendix, and every assumption is labelled.

2. EXECUTIVE SUMMARY

Situation. TEXIN sold 5.02 million sportswear pieces to LSC in 2026 at an ex-works (EXW) value of USD 19.28 million, with all process steam raised in a coal boiler that emitted 28,380 tCO2e. LSC has signalled potential orders growing 22%, 25% and 28% a year from 2027 to 2029 if TEXIN commits to a five-year ESG roadmap, and it imposes annual emission caps that tighten from 15% to 64% below the 2026 baseline by 2031.

Complication. The orders are visibility, not purchase orders, and they are released only after each year's ESG review. Because heat demand scales with output (0.0598 GJ per piece; see Appendix), winning the growth on coal would breach the very first cap: 31,786 t against 24,123 t in 2027. The equipment that actually burns biomass costs USD 830,500, or 88.3% of the USD 940,500 envelope, and delivers no reduction until all of it is commissioned. Each year must also clear a net profit margin (NPM) floor.

Question. How should TEXIN phase the USD 940,500 so that potential orders become profitable sales while every buyer cap and every P&L constraint is met from 2027 to 2031?

Answer. Front-load the investment. Tranche 1 (USD 896,500, 95.3% of the envelope) enters service on 1 April 2027 and Tranche 2 (condensate recovery, USD 44,000) on 1 March 2028. Table 1 summarises the outcome: the plan is compliant in all five years, self-funded, and every year of delay would cost about USD 20 million of revenue.

Table 1. Headline results of the recommended plan (policy A+H, 2027 to 2031)

Dimension	Result	Case requirement
ESG compliance	Net CO2e 8.6% to 13.2% below cap; 67.8% cut by 2031 (60.1% in-boundary)	64% cut by 2031; annual caps

Dimension	Result	Case requirement
Sales	Realized sales 5.62M, 6.92M, 8.65M pcs (2027 to 2029), then held at 8.65M; USD 147.9M revenue	Justified conversion and own 2030 to 2031 forecast
Profitability	NPM 9.50%, 9.29%, 9.21%, 9.12%, 9.02%; EXW price must fall 11.8% to breach a floor	≥ -10%, ≥ -8%, then > 0%
Cash flow	Free cash flow positive every year; USD 11.4M cumulative; no new debt	Fund CAPEX, premium, payback logic
Value of ESG	USD 69.6M revenue and USD 6.6M net profit at risk without ESG; a one-year delay costs USD 21.0M revenue	Quantify revenue at risk from delay

From 2030 the carbon cap, not demand, limits output. The recommended operating policy, A+H (Section 6), raises biomass to 82.5% and 85% and covers the rest of the reduction with 3,315 tCO2e of carbon credits, holding 8.65 million pieces: USD 13.5 million more revenue and USD 1.36 million more net profit than Option A. Credit use is confirmed with LSC at the 2029 checkpoint.

3. INVESTMENT ROADMAP

3.1 Why the roadmap must be front-loaded

The case allows the CAPEX envelope to be phased over five years but not reduced in scope. Two facts in the workbook decide how much freedom that really gives. First, the 2027 cap (24,123 tCO2e) is set by the buyer, not by TEXIN's investment calendar. Second, eight of the eleven items (boiler retrofit, feeding, storage, ash handling, filters, fire safety, civil works and engineering) are jointly required before any biomass can be fired safely and legally; none of them lowers emissions on its own. An even phasing of 20% a year would leave TEXIN on coal in 2027 and breach the cap by 31.8%, so it is not a feasible option. The real decision is therefore *when* the firing core enters service and *how hard* it is run. Only the condensate recovery upgrade is separable, which defines the two tranches below.

3.2 CAPEX tranches and commissioning schedule

Table 2. CAPEX by item and tranche: case cost plus the 10% contingency allocated to each item (USD)

CAPEX item	Case priority	Case cost	Tranche 1 (2027) +10% cont.	Tranche 2 (2028) +10% cont.	In service
Biomass boiler/furnace retrofit	Critical	280,000	308,000		Apr 2027
Fuel receiving, conveyor and feeding	Critical	120,000	132,000		Apr 2027
Covered warehouse and day silo	Critical	85,000	93,500		Mar 2027
Multi-cyclone and bag filter	Critical	95,000	104,500		Apr 2027

CAPEX item	Case priority	Case cost	Tranche 1 (2027) +10% cont.	Tranche 2 (2028) +10% cont.	In service
Fire detection and dust control	Critical	30,000	33,000		Mar 2027
Ash extraction and handling	High	45,000	49,500		Apr 2027
Civil, electrical and piping	High	55,000	60,500		Mar 2027
Engineering, commissioning and test	High	45,000	49,500		Apr 2027
Stack emission and energy metering	High	35,000	38,500		Jan 2027
ESG certification and traceability	Medium	25,000	27,500		Feb 2027
Water treatment and condensate recovery	Medium	40,000		44,000	Mar 2028
Total per tranche		**855,000**	**896,500**	**44,000**	
Share of envelope (USD 940,500)			95.3%	4.7%	
Annual depreciation (5-year SL)			179,300	8,800	

Note: each tranche value is the case cost × 1.10. The 10% contingency (USD 85,500, ESG_CAPEX OPEX B29) is allocated pro rata to the items so that the two tranches add up to the TOTAL PROJECT CAPEX envelope of USD 940,500 required by the case rules; depreciating only the USD 855,000 subtotal would understate it at USD 171,000 instead of USD 188,100 a year.

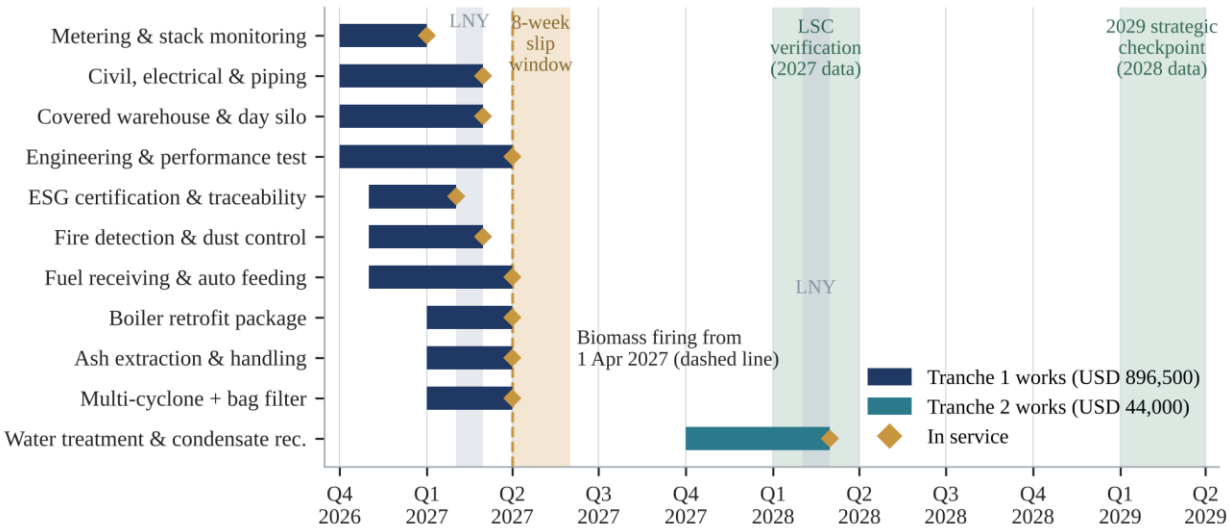


Figure 1. Commissioning schedule of the eleven CAPEX items and the buyer review milestones. *Source: ESG_CAPEX OPEX B17:B27; Emission Target B18:B19; team schedule.*

Works start in October 2026 so that metering is live from 1 January 2027 and 2027 performance is measured from the first day, which matches the case milestone "quick wins and metering". Hot works on the boiler are scheduled into the Lunar New Year shutdown of February 2027, and the performance test closes on 31 March, so biomass firing starts on **1 April 2027** (Figure 1). The schedule keeps an eight-week buffer: if commissioning slips to late May (risk R14), 2027 emissions rise to 24,029 t, still below the 24,123 t cap. Tranche 2 follows in the 2028 shutdown because it improves efficiency but is not needed for compliance, and sizing it on a year of biomass operating data avoids over-design.

3.3 Cumulative ESG completion and emission compliance

We define cumulative ESG completion as the value of CAPEX *in service* divided by the total project CAPEX, because the case states that a commitment alone does not count (Emission Target F18). Completion therefore moves in steps: 4.1% in January 2027 (metering), 26.9% in March, **95.3% from 1 April 2027** and 100% from March 2028 (Figure 3a).

Emissions are computed as heat demand times the emission factor of the fuel mix (coal 94.6 and biomass blend 9.1 kgCO2e/GJ), with heat scaling with output. Each year the boiler runs the lowest biomass share, in 5% steps, that keeps CO2e at least 5% below both the absolute cap and the intensity cap: 45%, 60% and then 80% while biomass is in service, or 33.9% as an annual average in 2027. Figure 2 shows that TEXIN complies in every year with a buffer of 6.1% to 13.2%, and the 2031 reduction reaches 66.2% against the 64% target.

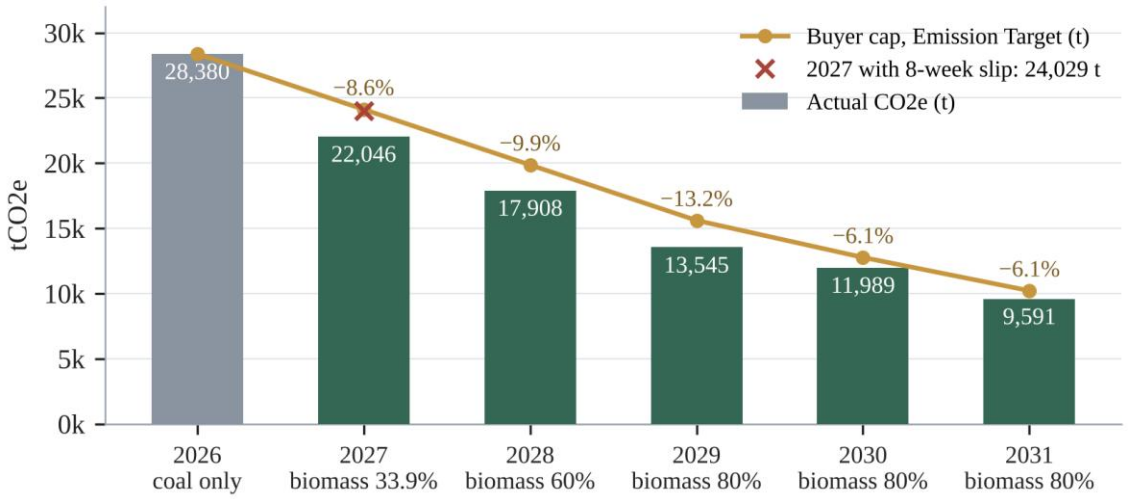


Figure 2. Annual CO2e against the buyer cap, with the operating biomass share. *Source: Emission Target D6:D11; Task 1 sheet.*

3.4 Depreciation by tranche

Following the case rule, each tranche is depreciated on a straight-line basis over five years from its commissioning year, with no salvage value. This replaces the component lives in the CAPEX sheet (USD 75,958 a year). Depreciation is USD 179,300 in 2027 and USD 188,100 from 2028; by the end of 2031 USD 931,700 has been expensed and only USD 8,800 of Tranche 2 remains on the books for 2032 (Figure 3b).

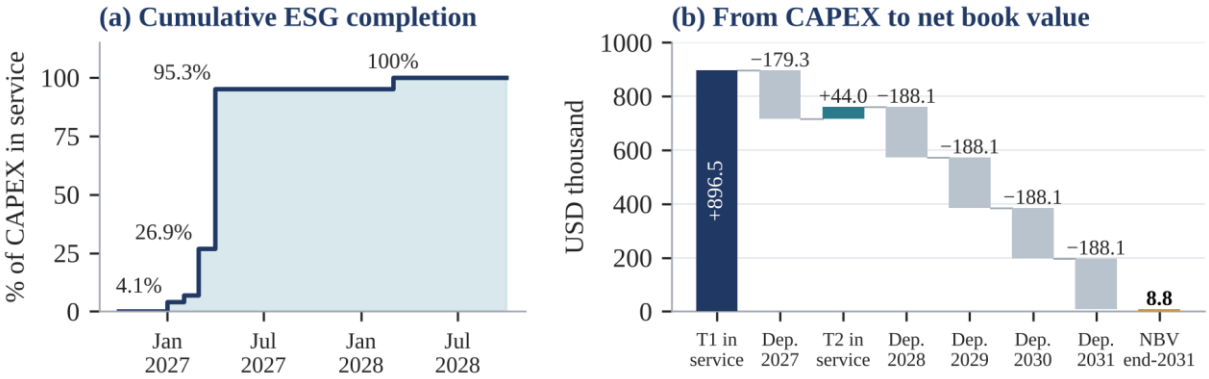


Figure 3. ESG completion (a) and CAPEX to net book value (b). *Source: Team calculation; case rule B27.*

4. SALES CONVERSION & DEMAND FORECASTING

4.1 Conversion of potential orders, 2027 to 2029

Realized sales are set by the tighter of two limits: risk-adjusted demand after LSC's ESG gate, and the output that keeps TEXIN within the buyer's CO2e caps. For demand we use the case's own reallocation rate instead of an arbitrary percentage. The Traditional scenario tells us how fast LSC moves orders away from a supplier whose ESG delivery is not proven (10%, 12% and 15% a year). Until each year's milestone is verified, that share of the potential order book is at risk, so

$$\mathbf{CR = (1 + g_{ESG} - |g_{Trad}|) \div (1 + g_{ESG})}$$

which gives 91.80%, 90.40% and 88.28% (Table 3). The rate falls while completion rises because each milestone is harder than the last (Moderate, Challenging, High) and the case itself warns of "accelerated order loss" in 2029. The capacity side does not bind in these years: at the 78% design mix the caps allow 9.10, 11.91 and 9.36 million pieces. We replaced the template's placeholder rates (90%, 94%, 97%), which rise over time and have no stated basis.

Table 3. Sales conversion rate by year and its justification

Year	ESG milestone delivered	Buyer challenge	Potential orders (pcs)	CR	Realized (pcs)
2027	Tranche 1 firing from 1 Apr; 95.3% complete	Moderate	6,124,400	91.80%	5,622,400
2028	100% complete; 2027 result verified	Challenging	7,655,500	90.40%	6,920,572
2029	Strategic checkpoint passed	High	9,799,040	88.28%	8,650,715

4.2 Independent forecast for 2030 and 2031

LSC gives no forecast for 2030 and 2031, and the case forbids extending the 22% to 28% growth mechanically. We split each scenario growth rate with the shift-share method (Dunn, 1960) into a market component, the mean of the ESG and Traditional rates (6.0%, 6.5%, 6.5%; average **6.33%**), and a competitive shift of 16% to 21.5% that captures reallocation toward compliant suppliers. After the 2029 checkpoint that reallocation is complete, so orders grow with the market only, to 10.42 and 11.08 million pieces, and the conversion rate is held at the

2029 level because the challenge stays High and Stretch. Risk-adjusted demand is 9.20 and 9.78 million pieces.

Two bounding scenarios test this (*assumption*): an upside with 6.5% market growth and the conversion rate back at the 2027 level of 91.8%, and a downside with 6.0% growth and the rate continuing its trend of minus 1.76 points a year (86.5% and 84.8%). Figure 4 shows the key result: in every scenario demand exceeds the carbon-allowed capacity, by at least 1.33 million pieces in 2030 and 3.21 million in 2031. Realized sales are therefore **7.66 and 6.13 million pieces** whichever demand case occurs. From 2030 the limit is carbon, not the market, so 2030 to 2031 growth becomes an operating decision (a higher biomass share or less heat per piece) that we assess in Section 6.

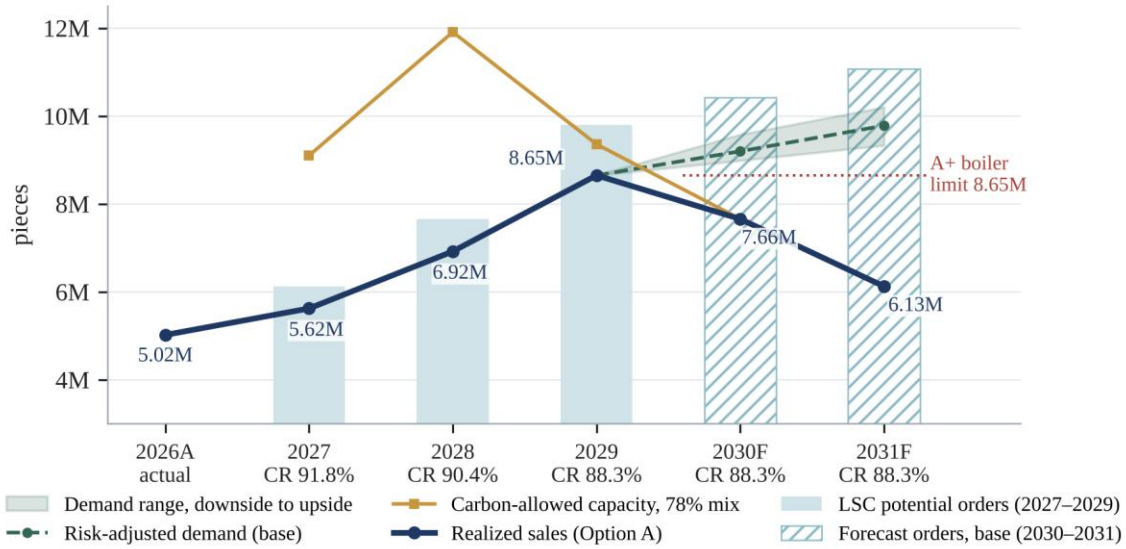


Figure 4. Potential orders, risk-adjusted demand and realized sales, with the 2030 to 2031 scenario band. *Source: Sales Forecast & Case 2A KPI B7:B12, D21:F21; Task 2 and Carbon-Allowed Capacity sheets.*

5. FINANCIAL & SCENARIO EVALUATION

5.1 Five-year P&L and constraint test

Revenue is realized volume times the weighted EXW price of USD 3.8415, which already contains the 1.5% to 2.5% ESG and traceability premium. Base COGS is USD 3.0525 per piece and already carries coal-fired energy, so we add only the *incremental* ESG OPEX: biomass fuel above the cost of coal for the same heat, biomass logistics, extra ash, handling labour (3 FTE per 18,125 t), maintenance above the coal level, monitoring, insurance and audit. At the case basis this reproduces the workbook's VND 14.83 billion a year. The case gives no overheads or tax, which would leave the NPM floors almost meaningless, so we add SG&A of 2.5% of revenue for selling plus G&A of 4.5% of 2026 revenue, half of it fixed. Listed peers report selling 2.9% and G&A 5.6% (May Song Hong, H1 2024) and total SG&A of 5.4% (TNG, Q4 2025); our selling ratio is lower because under EXW the buyer pays the main freight. Corporate income tax is 20% (Law No. 67/2025/QH15), FX is VND 25,500 per USD, and no new debt is needed.

Table 4. Five-year P&L of the ESG roadmap, base case (Option A, USD thousand)

USD thousand	2026A	2027	2028	2029	2030F	2031F
Realized sales ('000 pcs)	5,020	5,622	6,921	8,651	7,657	6,125
Revenue	**19,284**	**21,598**	**26,585**	**33,232**	**29,414**	**23,531**
Base COGS	(15,323)	(17,162)	(21,125)	(26,406)	(23,372)	(18,698)
Gross profit	3,961	4,436	5,460	6,825	6,041	4,833
Incremental ESG OPEX	-	(232)	(488)	(799)	(707)	(569)
Depreciation (ESG CAPEX)	-	(179)	(188)	(188)	(188)	(188)
SG&A	(1,350)	(1,460)	(1,697)	(2,012)	(1,831)	(1,552)
EBIT	2,611	2,565	3,088	3,826	3,315	2,525
CIT (20%)	(522)	(513)	(618)	(765)	(663)	(505)
Net profit	**2,089**	**2,052**	**2,470**	**3,061**	**2,652**	**2,020**
Net profit margin	**10.83%**	**9.50%**	**9.29%**	**9.21%**	**9.02%**	**8.58%**
Case NPM floor	-	≥ -10%	≥ -8%	> 0%	> 0%	> 0%
Constraint test	-	PASS	PASS	PASS	PASS	PASS
Free cash flow	-	974	1,837	2,213	3,436	3,125

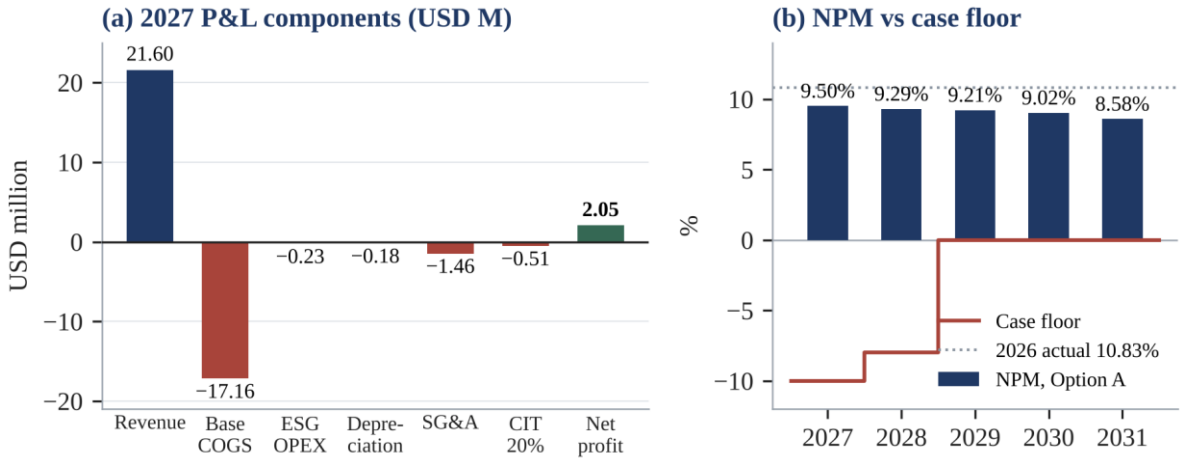


Figure 5. Profit bridge for 2027 (a) and net profit margin against the case floor (b). *Source: Task 3 sheet; Sales Forecast & Case 2A KPI B29:B32.*

Every year passes its floor with wide headroom: the weakest margin is 8.58% in 2031 against a floor of zero (Figure 5b). The ESG cost burden (incremental OPEX plus depreciation) is 1.9% of revenue in 2027 and rises to 3.2% in 2031 as the biomass share grows and volume falls, which explains most of the decline from the 10.83% of 2026. The 2027 bridge (Figure 5a) shows why the first year is safe: depreciation and ESG OPEX together take only USD 0.41 million out of USD 4.44 million of gross profit. Net profit plus depreciation in 2027 covers the Tranche 1 cheque 2.49 times, and free cash flow stays positive in every year after funding working capital, so the roadmap needs no new debt.

5.2 Sensitivity and break-even analysis

Figure 6 moves one driver at a time around the base case and records the change in five-year net profit; Table 5 gives the value at which the first NPM floor would be breached. Two findings matter for management. First, profit is far more exposed to commercial terms than to fuel or currency: losing the 1.9% ESG premium costs USD 1.95 million, almost twice the effect of the 28% rice husk price shock in R03. Second, conversion matters less than expected (±USD 0.30 million) because 2030 and 2031 volume is capped by carbon, not demand. None of the drivers comes close to a floor within its realistic range.

Table 5. Break-even values: change needed before the first NPM floor is breached

Driver	Base value	Break-even value	Required change	Reading
Weighted EXW price	USD 3.84/pc	USD 3.41/pc	-11.3%	6 times the ESG premium
Rice husk pellet price	VND 1.95M/t	VND 6.97M/t	+258%	9 times the R03 shock
SG&A ratio	7.0% of revenue	18.4%	×2.6	Far above peers
Conversion rate	88% to 92%	13% to 16%	-75.5 pts	Volume cut by over 80%
USD/VND rate	25,500	about 4,700	USD/VND -82%	Not plausible

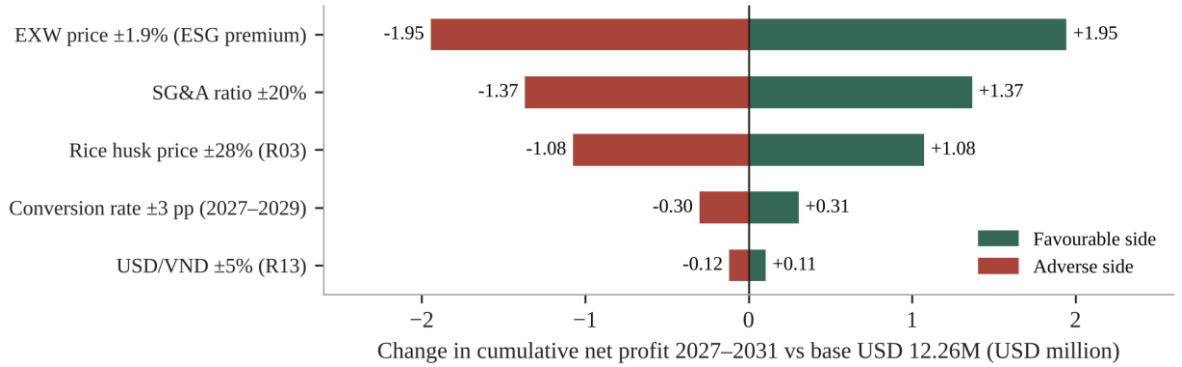


Figure 6. Tornado chart of five-year net profit (Option A). *Source: Team model; Supply Chain Risks C15, C25.*

5.3 ESG pathway versus Traditional and the cost of delay

The Traditional pathway keeps the coal boiler and sells at the base EXW price without the premium (USD 3.7698). Its volume falls 10%, 12% and 15% and, following the case's "accelerated order loss", a further 15% a year in 2030 and 2031. It is not a legal steady state: coal runs at 94.6 kgCO2e/GJ against an intensity cap of 80.41 in 2027 and lower afterwards. Even so, it serves as the counterfactual. Over 2027 to 2031 the ESG pathway earns **USD 69.6 million more revenue** (96% from volume, 4% from the premium) and USD 6.6 million more net profit, and its incremental free cash flow pays back in 2030 (Figure 7).

Delaying the roadmap is almost as costly per year as abandoning it (Table 6). Orders erode first under the Traditional rates, and the ramp then restarts from a lower base, so each year of delay

removes about USD 20 million of revenue, or roughly USD 1.7 million for every month. The loss sits in 2027 to 2030; by 2031 every converting path reaches the same carbon-allowed capacity. A delay also means one or two years in breach of the cap, which under the case rules lets LSC freeze or reallocate orders.

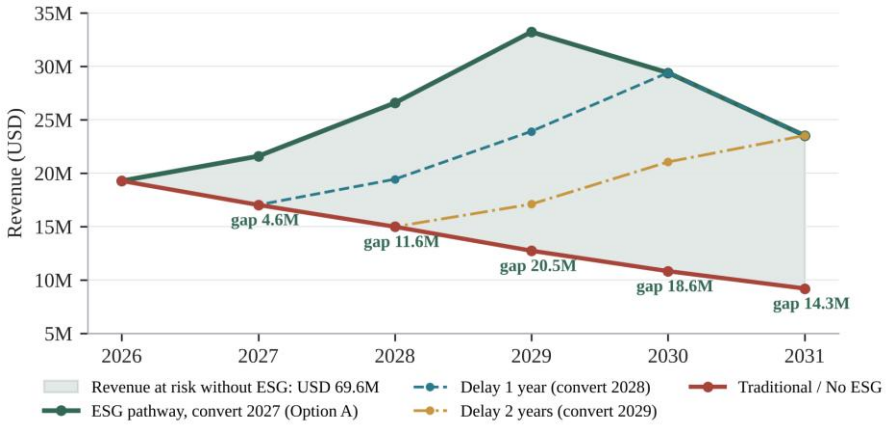


Figure 7. Revenue of the ESG pathway, the delay paths and the Traditional pathway. *Source: Task 4 sheet; EXW Price E6:G10.*

Table 6. Cost of delay, cumulative 2027 to 2031 versus converting in 2027 (USD million)

Measure	Delay 1 year	Delay 2 years	No ESG
Revenue at risk	**21.0**	**40.6**	**69.6**
Gross profit at risk	4.6	8.8	15.3
Net profit at risk	2.1	4.0	6.6
Years in breach of buyer cap	1 (2027)	2 (2027, 2028)	5 (all)
Revenue lost per month of delay	1.75	1.69	n/a

6. RECOMMENDATION

6.1 Phasing options, CRITIC weights and TOPSIS ranking

Credit assumptions (for investment phasing only). (1) TEXIN may buy carbon credits as quantified offsets (Emission Target B20): credits rated A to AAA at USD 14.80/t (Regreener, n.d.), expensed and tax-deductible. (2) Credits count toward a buyer cap up to 30% of that cap, as in Vietnam's carbon market (VnEconomy, 2026). Credits never change in-boundary CO2e (Emission Target B17), which stays a scored sub-criterion; Sections 3 to 5 use no credits.

All four options front-load the CAPEX, because delay or even phasing breaches the 2027 or 2028 cap; they differ only in 2030 to 2031, when the cap binds. **A** lets output fall to 6.13 million pieces; **A+** raises biomass to 85% and 90% to hold 8.65 million; **A+C** keeps the A mix and buys credits for the full A+ reduction (6,630 t); **A+H** does half the step-up (82.5%, 85%) and buys credits for the rest (3,315 t). Nine sub-criteria are normalised by min-max, weighted by CRITIC (Diakoulaki et al., 1995) and ranked by TOPSIS (Hwang & Yoon, 1981) in Table 7.

Table 7. CRITIC weights, weighted matrix V (4 × 9) and TOPSIS ranking

Criterion	Sub-criterion	B/C	w	A	A+	A+C	A+H
Growth	C1 Revenue	B	11.1%	0.000	0.111	0.111	0.111

Criterion	Sub-criterion	B/C	w	A	A+	A+C	A+H
	C2 2031 sales ÷ 2026	B	11.1%	0.000	0.111	0.111	0.111
Profitability	C3 Net profit	B	10.9%	0.000	0.107	0.109	0.108
	C4 Lowest NPM	B	10.5%	0.000	0.090	0.105	0.097
Cash flow	C5 Cumulative FCF	B	15.0%	0.150	0.000	0.029	0.015
	C6 Lowest annual FCF	B	0.0%	0.000	0.000	0.000	0.000
ESG	C7 Credit cost	C	13.6%	0.136	0.136	0.000	0.068
	C8 CO2e cut, in-boundary	B	12.6%	0.113	0.126	0.000	0.063
Execution risk	C9 Peak biomass tonnage	C	15.2%	0.152	0.000	0.152	0.076
TOPSIS closeness CC = D- ÷ (D+ + D-)				**0.5597**	**0.5661**	**0.5468**	**0.5750**
Passes all gates (credits ≤ 30% of cap)				Yes	Yes	No (43%)	**Yes**
Rank among eligible options				**3**	**2**	**n/a**	**1**

A+H ranks first (CC 0.5750), ahead of A+ (0.5661), A (0.5597) and A+C (0.5468), so it leads even before the offset gate that A+C fails (43% of the 2031 cap). It balances half the extra tonnage of A+ against half the credit bill of A+C and earns more than A+ while credits cost under USD 21.2/t. VIKOR agrees; a compensatory weighted sum prefers A+, so with a 0.009 margin the credit leg stays gated.

6.2 Recommended plan and decision gate

Phasing. Commission Tranche 1 (USD 896,500) for biomass firing from 1 April 2027 and Tranche 2 (USD 44,000) from 1 March 2028, funded from operating cash flow.

Operating policy. Run the minimum compliant mix to 2029, then A+H: 82.5% and 85% biomass plus 1,105 and 2,210 tCO2e of credits (USD 49,000), holding 8.65 million pieces with net CO2e at least 10.7% below the caps; credits are reported apart from in-boundary CO2e (R07). **Switch rule.** At the 2029 checkpoint TEXIN agrees offset use with LSC. If LSC refuses, or credits exceed USD 21.2/t or lack an A to AAA rating, TEXIN steps up to A+ (no new CAPEX) if supplier OTIF is at least 95%, three approved suppliers show chain-of-custody, moisture is 12% or less and boiler efficiency is at least 92% of the test; otherwise it stays on A.

Resilience. Under A+H, all six shocks at once (rice husk +28%, efficiency -8%, VND +5%, no premium, volume -10%, credits at USD 40/t) leave a minimum NPM of 5.35%, above every floor.

Commercial terms. Anchor the ESG premium, the largest profit driver, in the supply agreement against LSC's annual KPI pack, and index biomass prices so that fuel inflation is shared rather than absorbed.

6.3 Top 5 risks and mitigation

Of the fourteen register risks, thirteen are rated High. Under EXW the buyer controls main transport (EXW Price D18), so R09 to R11 stay on a watch list shared with LSC, and R03, R13 and R14 are already quantified in Sections 3 and 5. The Top 5 are the risks that can break the

biomass system or the credibility of its emission claims (Figure 8, Table 8); residual ratings are the team's assessment.

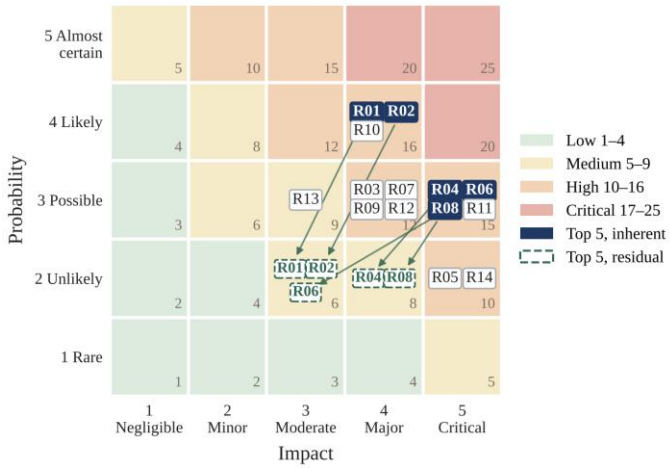


Figure 8. Risk heat map on the case scale (probability × impact) with Top 5 movement after mitigation. *Source: Supply Chain Risks A5:M26; residual scores are team assessment.*

Table 8. Top 5 risk register: trigger, response and owner

Risk (P×I)	Trigger / KPI	Preventive and contingency actions	Residual score	Owner
R01 Rice husk supply in rainy season (16)	Supplier OTIF < 85%	Three approved suppliers; 10 to 14 day covered buffer built before the rainy season; switch share to wood pellet	2×3 = 6	Procurement
R02 Moisture rises from 10% to 18% (16)	Moisture > 12% at receiving	Contract spec ≤ 10%; test every lot with a rejection rule; covered FIFO storage; blend with wood pellet	2×3 = 6	QA + Utilities
R04 Silica ash fouling (15)	Efficiency -8%; cleaning doubles	Redesigned preventive maintenance and cleaning cycle; critical spares; higher wood pellet share	2×4 = 8	Engineering
R06 Incomplete fuel traceability (15)	Lot without chain-of-custody	Lot-level chain-of-custody; annual supplier audit; lots without evidence excluded from biomass CO2e claims	2×3 = 6	ESG + Procurement
R08 Steam pressure affects dyeing (15)	Temperature deviation > spec	Steam pressure KPI linked to dyeing QC; load test at 1.72 times 2026 heat before the 2029 peak	2×4 = 8	Production

6.4 Implementation roadmap and KPI dashboard

Table 9 turns the recommendation into a calendar: every KPI has a target from the case or the stress tests, and every trigger names the action that follows.

Governance. Utilities, QA and Procurement check fuel and steam KPIs weekly; the monthly S&OP compares the CO2e run-rate with the pro-rata cap and sets next month's biomass share;

the board reviews the dashboard quarterly with the KPI pack that LSC verifies (Emission Target B18:B19). Each KPI has one named owner, as in Table 8.

Why these thresholds. A run-rate above 95% of the pro-rata cap means the 5% buffer is being used up; one 5% biomass step removes 4.3 kgCO2e per GJ, 9% to 15% of the 2028 to 2030 cap. A slip beyond four weeks is escalated because eight weeks is the last point at which the 2027 cap holds (24,029 t against 24,123 t). The credit leg and any switch to A+ need clear NPM headroom and credits below USD 21.2/t.

Evidence. All KPIs come from systems the CAPEX already funds (stack and energy meters, lot traceability, moisture testing at receiving), and the workbook recalculates the cap, biomass share and NPM for any new input, so the dashboard refreshes monthly.

Table 9. Implementation roadmap, KPIs and decision triggers

Window	Key actions	KPI and target	Trigger and response
Q4 2026	Sign retrofit contract and three index-linked fuel contracts; civil works; metering live	Metering live 1 Jan 2027	Any contract late > 4 weeks: escalate to the board
Q1 2027	Boiler works in the LNY shutdown; build 10 to 14 day covered buffer; performance test	Test passed by 31 Mar 2027 (R14)	Slip > 4 weeks: fast-track spares and crew; an 8-week slip still complies
Apr 2027 to 2028	Biomass at 45% then 60%; Tranche 2 in the 2028 shutdown; first LSC verification	CO2e run-rate at least 5% below pro-rata cap; OTIF ≥ 95%; moisture ≤ 10%	Run-rate > 95% of cap: raise biomass by one 5% step
2029	80% biomass at peak load (1.72 times 2026 heat); strategic checkpoint	Boiler efficiency ≥ 92% of test; steam in spec; NPM headroom ≥ 5 pts	Offsets OK, ≤ USD 21.2/t: A+H; else A+ or A
2030 to 2031	82.5% and 85% biomass plus 3,315 t of credits (A+H)	2031 cut ≥ 64% net; credits ≤ 30% of cap; full chain-of-custody	Credits > USD 21.2/t or refused: step up to A+

Together, the plan converts LSC's ESG demand into USD 147.9 million of revenue over five years, keeps TEXIN inside every cap and profit floor, and turns the 2030 to 2031 lever into a decision taken on evidence rather than on hope.

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APPENDIX. KEY ASSUMPTIONS AND RESOLUTION OF CASE DATA CONFLICTS

Table 10. How missing or conflicting case data were handled

Issue in the case data	Resolution adopted	Basis
Heat demand fixed at 300,000 GJ while volume almost doubles	Heat scales with output at 0.0598 GJ per piece	ESG_CAPEX OPEX B6, D6 (steam for dyeing and finishing)
"78% biomass gives 64% reduction", while the workbook factors give 70.5%	Workbook emission factors used; 78% kept only as the design mix for capacity	CO2 Emission Calculation; Emission Target F11
COGS already contains coal-based energy	Only incremental ESG OPEX above coal is added	COGS sheet; ESG_CAPEX OPEX F45
No SG&A, interest or tax given	SG&A 7.0% (half of G&A fixed); CIT 20%; no new debt	MSH and TNG reports; Law No. 67/2025/QH15
2030 to 2031 constraint worded differently in brief and workbook	Stricter version applied: net profit and NPM > 0%	Case brief Table 3; KPI sheet B32
No boiler rating given	Retrofit assumed to supply up to 1.72 times 2026 heat; A+ volume capped there	ESG_CAPEX OPEX J17; flagged as R08
No working capital terms	Team assumption: receivables 45, inventory 45, payables 30 days	Used for free cash flow only